

Clans, Guilds, and Markets: Apprenticeship Institutions and Growth in the Preindustrial Economy

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- **The Core Question:** Why did Western Europe achieve technological primacy and pull ahead of other world regions (like China, India, and the Middle East) in the centuries leading up to the Industrial Revolution?
- **The Hypothesis:** Superior institutions for the creation and dissemination of productive, tacit knowledge explain the European advantage.
- **The Mechanism:** The transmission of skills from the old to the young through *apprenticeship*.
- **The Institutional Divergence:** Regions relying on extended kinship systems (Clans) faced limits on knowledge dissemination, whereas Europe's shift to non-kinship organizations (Guilds, Markets) spurred rapid technological progress.

The Nature of Tacit Knowledge & Apprenticeship

- Historically, technological knowledge was almost entirely “tacit” (learned via observation, imitation, and practice).
- **Apprenticeship** was the primary mechanism for transferring this human capital.
- **The Contracting Problem:** The master-apprentice relationship is the archetypal incomplete contract.
 - The apprentice supplies labor; the master supplies training and room/board.
 - The quality of teaching is unobservable and non-verifiable by third parties ex-ante.
 - Creates a severe **moral hazard** problem: Masters have an incentive to use the apprentice for labor (κ) but shirk on the costly effort of teaching (δ).

Four Institutional Archetypes

To solve the moral hazard problem, societies historically relied on different contract enforcement institutions:

- ① **The Family:** No external enforcement. Knowledge is transmitted strictly from parent to child.
- ② **The Clan:** Informal enforcement via trust and reputation within an extended kinship network.
- ③ **The Guild:** Semi-formal enforcement by a coalition of masters in a specific trade (crosses bloodlines, but acts as a cartel).
- ④ **The Market:** Formal third-party (state) enforcement preventing anticompetitive behavior.

- **Malthusian Economy**, all decisions are made by adults with utility

$$u(c, l') = c + \gamma l'$$

- **Aggregate Production:** Cobb-Douglas using land (X) and effective craftsmen labor (L):

$$Y = L^{1-\alpha} X^\alpha \quad \text{where} \quad \alpha \in (0, 1) \text{ and } X = 1$$

- **Effective Labor (L):** CES aggregate of labor supplied in different trades j :

$$L = \left(\int_0^1 (L_j)^{\frac{1}{\lambda}} dj \right)^\lambda \quad \text{where} \quad \lambda > 1$$

- **Malthusian Constraint:** Positive feedback from income per capita (y) to population survival (s):

$$n = \bar{n} \min[1, sy] \implies N' = nN$$

- **Productivity:** Craftsmen are heterogeneous. Output q_i for a craftsman with cost parameter h_i is $q_i = h_i^{-\theta}$.
- **Knowledge Distribution:** Cost parameter h_i follows an exponential distribution with rate k : $h_i \sim \text{Exp}(k)$.
- A higher k means lower average cost and higher average productivity.

- **Income per Capita:**

$$y = \frac{Y}{N} = \Gamma(1 - \theta)^{1-\alpha} k^{(1-\alpha)\theta} N^{-\alpha}$$

- **Balanced Growth Path (BGP):** To maintain constant income per capita y , productivity growth (g) and population growth (n) must satisfy:

$$g^{\theta(1-\alpha)} = n^{\alpha}$$

How does knowledge k evolve across generations?

- **Learning:** An apprentice learns from m masters. Because quality is unobservable, they adopt the technique of the best master they are exposed to:

$$h_L = \min\{h_1, h_2, \dots, h_m\}$$

- **Innovation:** Craftsmen generate new ideas $h_N \sim \text{Exp}(\nu k)$.
- **Final Knowledge:** $h' = \min\{h_L, h_N\}$.
- **Law of Motion for Knowledge:** With m randomly chosen masters, aggregate knowledge grows by:

$$k' = (m + \nu)k \implies g = m + \nu$$

The Cost of Apprenticeship

- An apprentice produces $\kappa > 0$ units of output for the master.
- To teach a apprentices, a master incurs a convex cost $\delta(a) = \frac{\bar{\delta}}{2}a^2$.
- **Moral Hazard:** A master can take a apprentices, collect κa , and provide no training (saving $\delta(a)$).
- Without institutional enforcement to punish shirking, parents will not send children to masters outside the family.

Institution 1: The Family Equilibrium

- **Setting:** Modeled as a game where craftsmen decide on training, exploiting, and punishing. No communication network exists to enforce contracts outside the immediate family.
- **Nash Strategy Profile:**
 - All craftsmen train only their own children.
 - *Off-path:* If a master gets someone else's child as an apprentice, they exploit them.
 - No one ever punishes anyone.
- **Growth:** Parental altruism ensures parents do not shirk on their own kin. Growth relies entirely on internal innovation (ν):

$$g^F = 1 + \nu$$

- **Implication:** Slowest rate of technological progress. If $\nu = 0$, the economy stagnates completely.

Institution 2: The Clan Equilibrium

- **Setting:** Decentralized enforcement is possible within an extended family/clan (size parameterized by o , generations of shared ancestry).
- **Nash Strategy Profile:**
 - Craftsmen send children to be trained by all clan masters, and masters put effort into teaching.
 - *Off-path:* If a master cheats an apprentice, all current members of the clan punish the master.
- **Limitation (Correlated Knowledge):** Because all current masters in a clan learned from the same ancestors, their knowledge is highly correlated.
- **Growth:** Faster than the family due to shared new ideas ($m^C = (n^C)^o$), but limited by clan size:

$$g^C = 1 + \frac{\nu(n^C)^o}{g^C - \nu}$$

- If innovation $\nu = 0$, growth is zero despite $m > 1$, because no new knowledge enters the closed loop.

Institution 3: The Guild Equilibrium

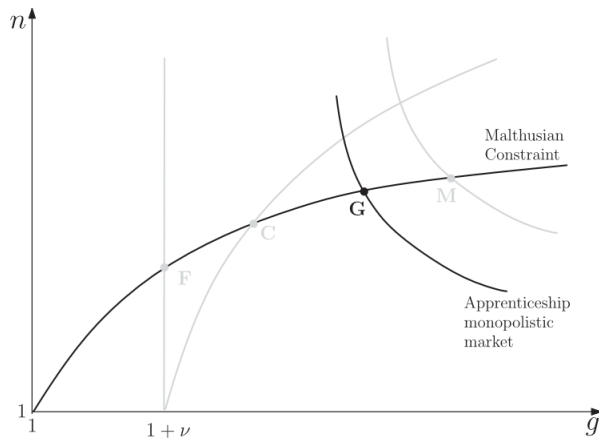
- **Setting:** A coalition of all masters in a trade. Unlike the decentralized Nash punishment games, guilds provide centralized enforcement across bloodlines, unlocking the full distribution of knowledge.
- **The Guild's Problem:** Instead of relying on decentralized punishment, the guild maximizes the utility of its masters by setting the price of apprenticeship and limiting supply (a).
- **Trade-off:**
 - *Pro:* Apprentices get independent draws of knowledge (uncorrelated), maximizing learning potential.
 - *Con:* Guilds act as monopolies to extract rents, limiting the quantity of training.
- **Growth:** The rate of knowledge growth $g^G = m^G + \nu$ depends on the guild's market power.
- If guilds' anticompetitive power is not overwhelming, the benefit of sampling uncorrelated knowledge vastly outweighs the clan's closed system.

Institution 4: The Market Equilibrium

- **Setting:** Centralized formal contract enforcement (e.g., state courts) replaces decentralized punishment networks.
- **Market Behavior:** Modeled as a complete-markets economy where contracts are perfectly enforced, eliminating the moral hazard (shirking) problem.
- **Outcome:** Unlike guilds, there is no anticompetitive behavior. Parents and masters act as price-takers in a competitive apprenticeship market.
- Masters solve a straightforward profit maximization problem, equalizing the price of apprenticeship to the marginal cost of training: $p = \delta'(a) - \kappa$.
- **Growth:** Apprentices get independent draws from the full knowledge distribution without monopoly restrictions. This yields the highest number of masters (m^M) and the fastest productivity growth:

$$g^M = m^M + \nu$$

Graphical Representation of Equilibria



Comparing Guilds vs. Clans (The Core Argument)

- Why did Europe (Guilds) overtake China (Clans)?
- Suppose a Guild limits apprenticeship such that m (number of masters per apprentice) is exactly the same as in a Clan.
- **The Guild still generates vastly more growth.**
- *Why?* In a Guild, masters are independent draws from the aggregate knowledge distribution. In a Clan, masters' knowledge is correlated (they all learned from the same grand-master).
- European institutions (journeyman-ship, tramping) allowed young artisans to sample diverse, uncorrelated techniques across regions.

Quantitative Assessment: Parameterizing the Model

- Model calibrated to historical data, with one period equal to 25 years.
- **Macro & Technology Parameters:**
 - Land share $\alpha = 1/3$.
 - Substitution $\lambda = 1.3$ (implies a 30% guild markup).
 - Knowledge dispersion $\theta = 0.5$.
- **Growth & Demographic Parameters:**
 - Innovation $\nu = 0.0086$ (matches earlier historical 0.86% population growth).
 - Clan size $o = 6$ (allows learning from distant relatives).
 - Apprentice output $\kappa = 0.02$ (In families, 1/3 as productive as adults).
- **The Core Experiment:** Training cost ($\bar{\delta} = 0.019$) is set to equalize the number of masters (m) in both Clan and Guild equilibria.
- *Purpose:* This isolates the growth effect driven purely by the **uncorrelated** knowledge available in Guilds.

Table I: Balanced Growth Paths in the Parameterized Economy

Equilibrium	Number of masters	Productivity growth	Population growth
Family F	1.00	0.29	0.86
Clan C	1.06	0.30	0.91
Guild G	1.06	2.14	6.43
Market M	1.08	2.87	8.60

Notes: One period corresponds to 25 years. Growth rates are in percent per period. Notice the massive jump in productivity from Clan to Guild, despite holding m at 1.06.

Table II: Accounting for Differences Between W. Europe and China

Can the model match the empirical gap in TFP growth between Europe (Guilds) and China (Clans)?

Period	Gap in TFP growth	Training cost ($\bar{\delta}$)	Number of masters
1000–1500	2.5 (Maddison)	0.016	1.15
1000–1500	4.8 (Broadberry et al.)	0.013	1.29
1500–1850	2.4 (Maddison)	0.018	1.07
1500–1850	7.4 (Broadberry et al.)	0.014	1.22

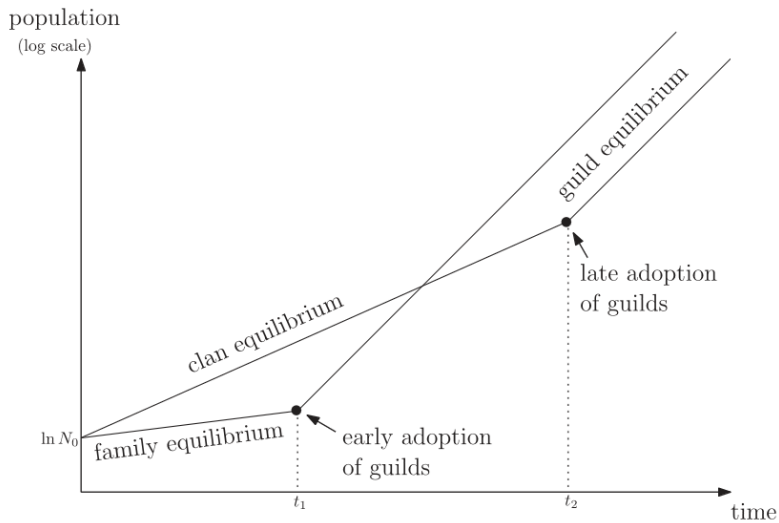
Conclusion: The observed gap can be matched with a very low number of masters per apprentice in the Guild system (between 1.07 and 1.29), making the mechanism highly plausible.

Endogenous Institutional Transitions

- **Question:** If Guilds are better, why didn't Clan-based societies adopt them?
- **Theory:** Adopting Guilds requires paying a fixed cost $\mu(N)$, which decreases with population density ($\mu'(N) < 0$).
- The incentive to pay this cost depends on the performance of the *status quo* institution.
- **The Clan's Curse:** The Clan equilibrium provides "good enough" growth and social insurance. The marginal benefit of adopting Guilds is small.
- **Europe's Starting Point:** Europe started in the low-growth Family equilibrium. The pain of stagnation made the fixed cost $\mu(N)$ worth paying much earlier.

The "Absorbing State" of Clans

- **Proposition 7:** If the fixed cost $\mu(N)$ does not strictly go to zero as population goes to infinity, an economy in the Clan equilibrium may **never** adopt the Guild.
- Let two economies start with the same low population N_0 .
- The Clan economy grows faster initially.
- However, the Family economy reaches its (lower) adoption threshold first, switches to Guilds, and accelerates past the Clan economy.
- The Clan economy gets trapped in a local optimum.



- The dissemination of tacit knowledge was the primary engine of preindustrial growth.
- Apprenticeship was the vehicle, but its success depended on the institutions enforcing it.
- **Key Insight:** European guilds facilitated knowledge transfer across bloodlines. Young apprentices were exposed to diverse, uncorrelated ideas, driving rapid technological drift.
- Europe's lack of strong kinship networks (Clans) forced it to develop corporate institutions (Guilds, Markets) that proved vastly superior for human capital accumulation.